Non-generic ECDLP leads for prime, binary and Koblitz curves

Search round, 2026-10-18 · study `research/ecdlp_leads_20261018` · repository `aburan28/crypto`

Status. Proposals and derivations only. Nothing in this report is a measurement, and no security claim about any deployed curve is made or withdrawn. Labels: *sourced* (cited fact), *derived* (follows by the argument shown; prose, no `IDC1` certificate yet), *computed* (shell arithmetic of a stated formula; data in `cover-genus-budget.csv`), *proposed* (open). The Markdown `README.md` beside this PDF is the full text.

1. Question and scope

The owner asked for 10 to 33 new leads for *non-generic* ECDLP algorithms over prime-field, binary and Koblitz curves. A lead is admitted only if it passes three tests. It is not already on the repository's list. It names a mechanism that the Corrigan-Gibbs–Kogan bound for generic algorithms does not cover. It has a first experiment and a kill criterion. Ten leads passed. Ten ideas were closed on paper while drafting.

2. Bottom line

1. **Prime fields: one real lever, and it is curve-specific.** On curves with small integer coefficients the summation polynomial S_3 has small integer coefficients. Its small-root lattice collapses from 6 unknown monomials to 3. With projective (Farey) factor bases the 3 unknowns are also balanced in size. The level-1 reach for $m = 2$ rises from $p^{1/12}$ to $p^{1/9}$ to $p^{1/6}$ (derived). This applies to secp256k1, Curve25519, Curve448, BN254 and BLS12-381 G1. The NIST P-curves ($a = -3$) get $p^{1/10}$. Random-coefficient curves get nothing. Since $m = 2$ needs $1/2$ and $m = 4$ needs about $1/5$, this is an oracle improvement, not a threat.
2. **Koblitz curves: the cover route is not dead.** GHS is degenerate on Koblitz curves. Two facts reopen it. First, the Klein quartic over $\mathbb{F}_2$ has $L(X) = 1 + 5X^3 + 8X^6$, which equals $L\big(\mathrm{Res}_{\mathbb{F}_8/\mathbb{F}_2} E_{a=1}\big)$. Second, Eichler–Shimura applied to the CM newform of 49a1 twisted by a character χ of order n with $\chi(2) \neq 1$ gives a reduction isogenous to the trace-zero variety T_n, for every prime n. Class-group index calculus over $\mathbb{F}_2$ is subexponential. A covering curve of genus at most $c(n) \cdot n$ beats rho, where $c(131) \approx 1.9$ and $c(571) \approx 6.1$ (computed).
3. **Random-b binary curves: only the cover question is new here.** A random-b curve does not lift to a Q-curve, so the Koblitz mechanisms do not transfer (derived sketch).

3. The ten leads

#	family	lead	status	first experiment	kill criterion
L8	Koblitz	Étale route: $D/\mathbb{F}_2$ with an order-n automorphism and an $\mathbb{F}_2$-map to E; its twist contains $E \times T_n$, genus $1 + n(g_0 - 1)$	proposed	enumerate genus-2 and genus-3 $B/\mathbb{F}_2$ covering E with $n \mid \#J_{B(\mathbb{F}_2)}$; test $L(B, \chi, X)$ for the factor $L_{E(X)}$	no hit for $g_0 \leq 3$
L6	Koblitz	Modular twists: $(f_E \otimes \chi) \bmod 2 \sim T_n$ for every n	derived	smallest curve containing $A_{f \otimes \chi}$ for $n = 3, 5, 7, 11$	genus grows faster than $c(n)n$
L10	binary	Genus-n Jacobians realising $\mathrm{Res}(E)$, beyond GHS	proposed	all genus-3 curves over $\mathbb{F}_2$, which traces t_3; genus-5 search for point counts $3, 5, 9, 17, 33 - 5t_5$	no genus-5 realisation
L9	Koblitz and binary	Certified lower bounds on the cover genus	derived (partial)	Howe's test on T_n; many-points (32, 5); point-count positivity	(produces bounds)
L1	prime	Coefficient collapse plus Farey bases	derived	Probes B/C with the collapsed and Farey lattices, $m = 2, 3, 4$, against random-coefficient controls	gain under $1.25\times$ at $m = 4$
L2	prime	Eisenstein–Farey bases for $j = 0$	derived	same harness, secp256k1-shaped toys	invariance lost under shifts

L7	Koblitz	Klein family: $\mathbb{Z}/n$ covers of E ramified at one place of degree $\mathrm{ord}_n(2)$	derived ($n = 3$)	$L(E, \chi, X)$ for $n = 7, 31, 73, 127$	no hit at 7 or 31
L3	prime	Algebraic-exposure ledger, 176 prime curves	proposed	table of super-monomial count × model × shape	none (audit)
L4	prime	Audit four unchecked prime-field IC papers	proposed	native re-implementation under matched rho	none below baseline
L5	prime	Global twist-collision relations	proposed	squarefree twist classes of $f(x)$, $x \le B$; ranks	yield below $B^{0.6}$

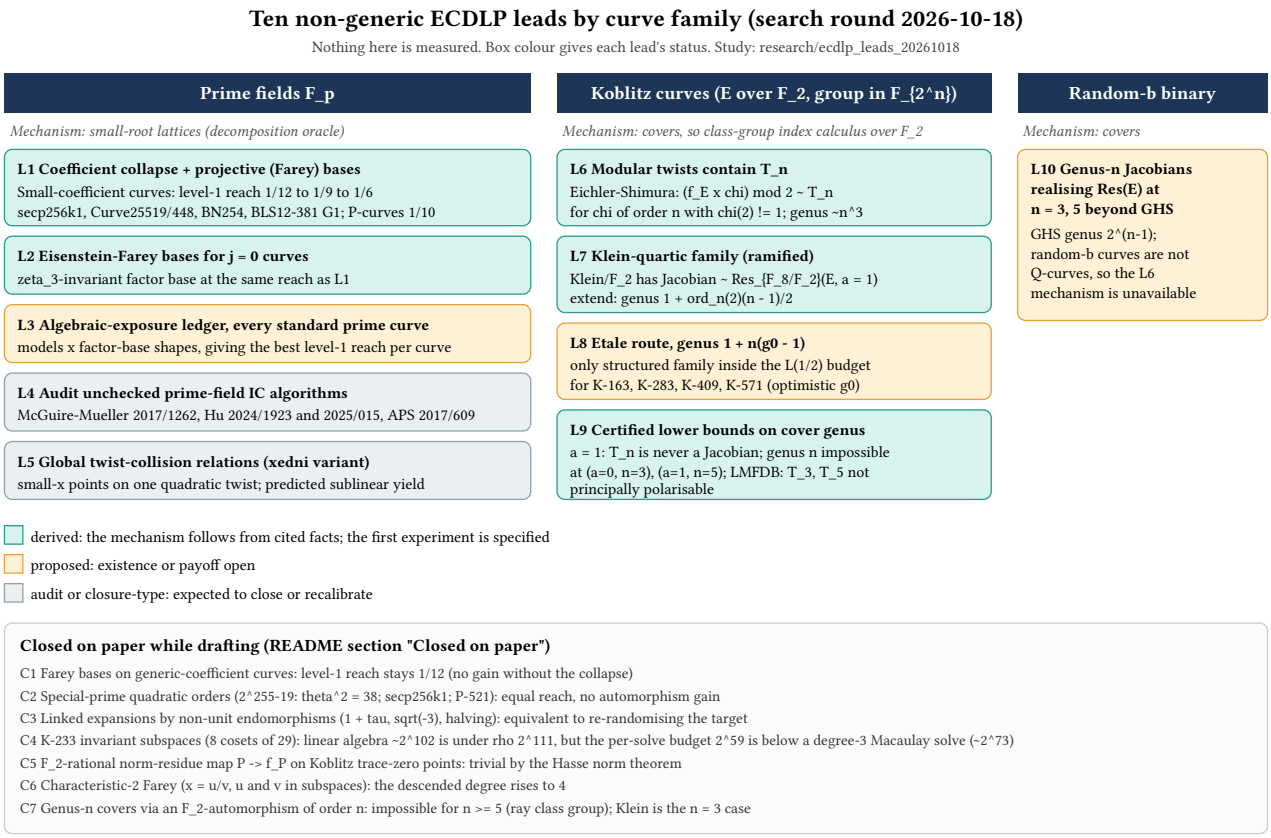


Figure 1: The ten leads by family with status colours, and the closures C1–C7. A diagram, not data.

4. Key derivations

L1, level-1 reach. $S_3(x_1, x_2, x_R) \equiv 0 \bmod p$ is a linear relation among unknown integer polynomials in the small summands. Reach is the largest factor base for which the product of their size bounds stays below p.

curve class	interval base	Farey base
generic a, b (5 unknowns plus a constant; Farey adds $s_1^2 s_2^2$)	$1/12$	$1/12$
$a = -3$, random b (P-256, P-384, P-521)	$1/10$	$1/10$
small a, b or small Montgomery A (3 unknowns c_2, c_1, c_0)	$1/9$	$1/6$
needed for decompositions to exist at $m = 2, 3, 4$	$1/2, 1/3, 1/4$	

For the collapsible class, $S_3 = c_2 x_R^2 - 2c_1 x_R + c_0$ with $c_2 = (x_1 - x_2)^2$, $c_1 = (x_1 + x_2)x_1 x_2 + 2b$ and $c_0 = (x_1 x_2)^2 - 4b(x_1 + x_2)$, all small integers. The Montgomery form is analogous with A. With $x = r/s$, each c_k becomes bihomogeneous of bidegree $(2, 2)$ and of size H^4. That gives $H^{12} \le p$ and $|F| \approx H^2 \le p^{1/6}$.

L6, Eichler–Shimura. $f_E \otimes \chi$ has level $49\ell^2$, character χ^2 and $a_2 = \chi(2)t$. At $p = 2$, $\mathrm{Frob}^2 - \chi(2)t\ \mathrm{Frob} + 2\chi(2)^2 = 0$, so $\mathrm{Frob} = \chi(2)\pi_E$. Over the conjugates these are exactly the eigenvalues of T_n. By Tate, $A_{f\otimes\chi} \bmod 2 \sim T_n$.

L7, Klein. The point counts are 3, 5, 24 over $\mathbb{F}_2, \mathbb{F}_4, \mathbb{F}_8$, so $L_{\mathrm{Klein}} = 1 + 5X^3 + 8X^6$. For the $a = 1$ Koblitz curve $t_3 = -5$, so $L(\mathrm{Res}) = 1 - t_3 X^3 + 8X^6$. These are equal, so by Tate the two varieties are isogenous.

L8, L9, closure C7. Tame $\mathbb{Z}/n$ ramification over $\mathbb{F}_2$ needs branch places of degree at least $\mathrm{ord}_n(2)$. Riemann–Hurwitz then leaves three options: quotient genus 0 or 1 (genus at least about $\mathrm{ord}_n(2)(n-1)/2$), or étale over a quotient of genus $g_0 \geq 2$ (genus $1 + n(g_0 - 1)$, needing $n \mid \#J_{B(\mathbb{F}_2)}$). Genus exactly n forces $n = 3$. If $J_C \sim T_n$ then $\#C(\mathbb{F}_{2^k}) = 2^k + 1 + t_k$ for $n \nmid k$. For $a = 1$ that gives 4 points over $\mathbb{F}_2$ but 2 over $\mathbb{F}_4$, which is impossible.

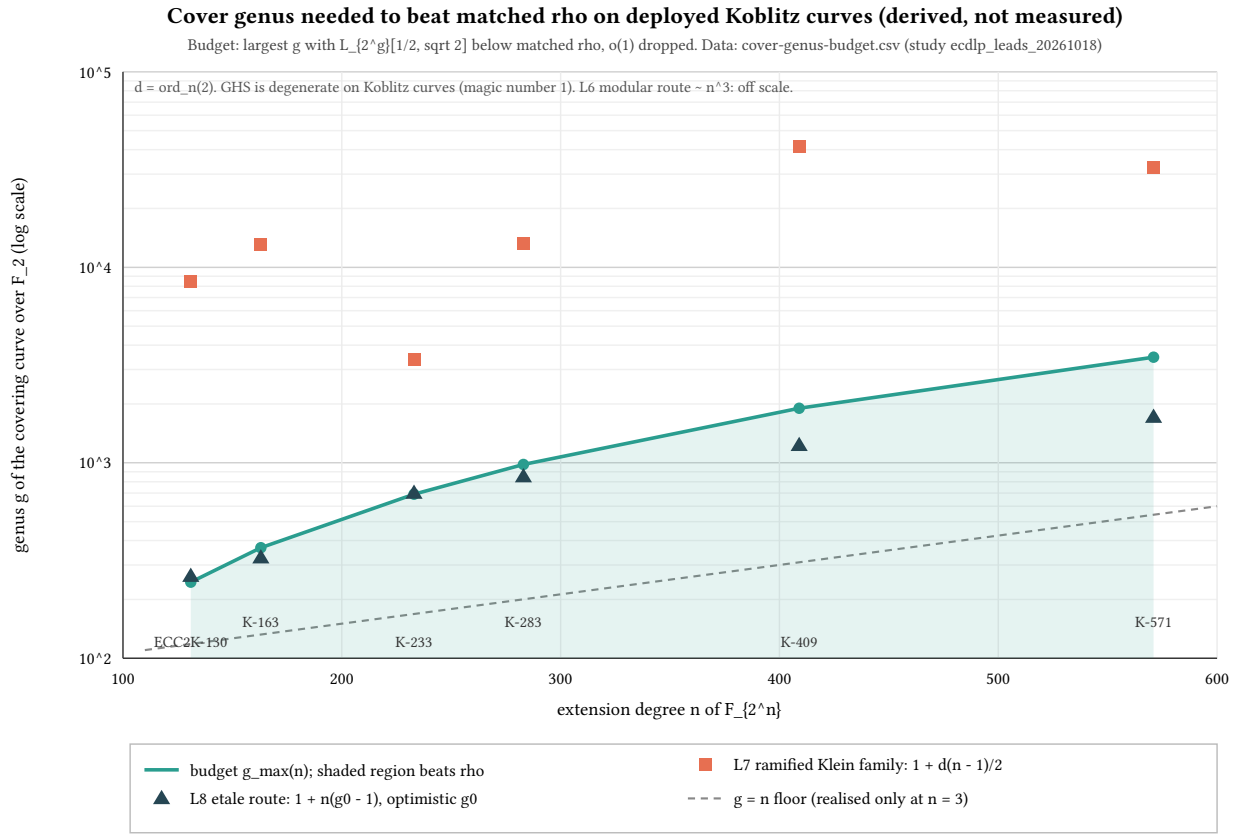


Figure 2: Derived, not measured. For each deployed Koblitz size: the L(1/2) genus budget $g_{\max}(n)$ (Enge–Gaudry constant $\sqrt{2}$, o(1) dropped), the étale-route genus $1 + n(g_0 - 1)$ at the smallest g_0 the Weil bound allows (optimistic), and the ramified Klein-family genus $1 + \mathrm{ord}_n(2)(n-1)/2$. Data: `cover-genus-budget.csv`.

5. Closed on paper while drafting

C1	Farey bases on generic curves	the constant becomes a sixth unknown; reach stays $1/12$
C2	Special-prime quadratic orders ($2^{255} - 19$, secp256k1, P-521)	equal reach with the collapse, worse without it; no automorphism gain
C3	Linked expansions by $1 + \tau$, $\sqrt{-3}$, halving	equivalent to re-randomising the target; towers raise the solving degree
C4	K-233 invariant subspaces (8 cosets of 29)	linear algebra $\approx 2^{102}$ is under rho $2^{110.9}$, but the per-solve budget 2^{59} is below a degree-3 Macaulay solve ($\approx 2^{73}$)
C5	K-571 and B-571 with $m = 5$ chained systems	needs solving degree at most 6 to 9 (the Semaev 2015 regime); the measured D growth contradicts it
C6	Oakley groups 3 and 4 by direct decomposition	per-trial Gröbner cost dominates; GHS stays the better route
C7	Genus-n covers with an $\mathbb{F}_2$-automorphism of order n	only $n = 3$ (Klein)
C8	Norm-residue map $P \mapsto f_P$ on Koblitz trace-zero points	trivial by the Hasse norm theorem
C9	Characteristic-2 Farey bases	the descended degree rises to 4
C10	Characters, Legendre symbols, pairings modulo small ℓ	these see only cofactors

6. Unresolved checks

- `identity.certificate/v1` records are owed for four claims: the L1/L2 counts, the Klein L-polynomial identity, the Eichler–Shimura twist, and the $a = 1$ obstruction.
- LMFDB returned nine classes. T_3 and T_5, for $a = 0$ and $a = 1$, are simple, not principally polarisable, and contain no Jacobian. At $n = 3$, $\mathrm{Res}_{\mathbb{F}_8/\mathbb{F}_2}(E)$ contains a genus-3 Jacobian exactly for $t_3 \in \{-5, -3, -1, 1\}$, the traces the point counts allow. The class for $t_3 = -5$ ($a = 1$ Koblitz) contains one Jacobian, the Klein quartic. So $n = 3$ cannot separate covers from GHS, and L10 starts at $n = 5$.
- Still blocked by a captcha: the nine admissible $n = 5$ classes, the two $n = 7$ trace-zero classes, and $t_3 = 5$ (already excluded by point counts).
- The manypoints bound for $(q, g) = (32, 5)$ is unchecked. It decides whether $(a = 0, n = 5)$ can be realised at genus 5.
- The novelty audit against the cover-attack literature (Diem, Scholten, Hess, Ekedahl–Serre, Elkies) is owed before any of L6–L10 is called new.
- The L(1/2) budget is optimistic: it uses the hyperelliptic constant and drops o(1). General curves need slower class-group algorithms.
- Graphs checked and left unchanged: the IC scoreboard, the progress timeline, the IC leaderboard and the ECDLP attack matrix. Nothing in this round is measured or certified.